JAX-sklearn (xlearn) — Drop-in sklearn with automatic JAX acceleration

github.com/chenxingqiang/jax-sklearn · 13,058 tests passed · pip install jax-sklearn

1. User code (zero API change)

import xlearn as sklearn
LinearRegression().fit(X, y)

Same estimators
cluster / decompose / preprocess

config_context()
force GPU / disable JAX

2. sklearn-compatible surface

Linear
LR / Ridge / Lasso
ElasticNet

Cluster
KMeans
282 tests

Decompose
PCA / TruncatedSVD
528 tests

Preprocess
Standard / MinMax

Others
native sklearn
via fallback

3. Runtime injection / proxy

__getattr__ lazy import
xlearn.linear_model …

JAX

_auto_jax_accelerate
replace estimator classes

EstimatorProxy
create JAX or original impl

Heuristics
fallback3 — KMeans > 1e6
PCA > 1e7 complexity

4. Acceleration + data

JAX kernels
JIT / vmap
universal mixins

Algo kernels
GP / MLP / SVM / tree

NumPy ↔ JAX
dtype / shape guards

Fallback
small data or error
→ original sklearn

5. Hardware abstraction (auto-select by problem size)

CPU
<10K samples
1.0–2.5×

NVIDIA GPU
10K–100K
5.5–8×

Google TPU
>100K
9.5–15×

Apple Metal
M1/M2/M3 beta
2–4×

Proxy decides JAX vs sklearn from dataset size, algorithm family, and configured platform; errors fall back silently.